INCH-POUND

MIL-P-82816(OS)
5 February 1990
SUPERSEDING
WS 22169
26 October 1983
(See 6.7)

MILITARY SPECIFICATION

PRIMER, ADHESIVE

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for an adhesive primer consisting of a coupling agent and a liquid vehicle, referred to herein as the primer.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-A-82845 Adhesive, Paste, One Part, Non-asbestos

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization Branch (3730), Indian Head, MD 20640-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 8040

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STANDARDS

FEDERAL

FED-STD-313 Material Safety Data, Transportation Data and Disposal Data for Hazardous Materials Furnished to Government Activities

MILITARY

MIL-STD-129 Marking for Shipment and Storage

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from: Military Specifications and Standards, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094).

2.1.2 **Other Government documents, drawings, and publications.** The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

CODE OF FEDERAL REGULATIONS (CFR)

49 CFR 100-199 Transportation

(Application for copies of CFRs should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402-0001.)

NAVAL SEA SYSTEMS COMMAND (CAGE Code 10001)

WR 43 Preparation of Quality Assurance Provisions (Including Classification of Characteristics)

(Copies of WR-43 are available from the Commanding Officer, Naval Publications and Forms Center, Attn: 1053, 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 **Non-Government publications.** The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN TRUCKING ASSOCIATION, INC.

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Association, Inc., Attn: Traffic Dept., 2200 Mill Road, Alexandria, VA 22314-4677.)

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NATIONAL RAILROAD FREIGHT COMMITTEE

Uniform Freight Classification (UFC) 6000

(Application for copies should be addressed to the National Railroad Freight Committee, 222 South Riverside Plaza, Suite 1120, Chicago, IL 60606-5945.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3762 Adhesive-Bonded Surface Durability of Aluminum (Wedge Test)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 **Order of precedence.** In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 **First article.** When specified (see 6.2), a sample of the primer shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.2 **Material.** The primer shall be a two-part system consisting of a coupling agent and a liquid vehicle.

3.2.1 **Coupling agent.** The coupling agent shall be Union Carbide A-187, or equal (see 6.4).

3.2.2 **Liquid vehicle.** The liquid vehicle shall consist of the following reagent grade chemicals:

a. Absolute ethanol
b. Distilled water
c. Toluene
d. N-butanol
e. Butyl cellosolve (2-Butoxyethanol)
f. Acetic acid

3.2.3 **Composition.** The primer composition shall be as specified in table I.

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TABLE I. Primer composition.

Ingredient	Parts by Weight
A-187, or equal	5.0 ± 0.1
Absolute ethanol	40 ± 1
Distilled water	5.0 ± 0.1
Toluene	40 ± 1
N-Butanol	5.0 ± 0.1
Butyl cellosolve	5.0 ± 0.1
Acetic acid	As required for pH of 4.0 to 5.0

3.2.4 **Mixing**. The primer ingredients shall be mixed in accordance with the following:

a. Combine all ingredients except the acetic acid in a clean, dry glass container or bottle and agitate until thoroughly mixed.

b. While stirring the mixture, slowly add acetic acid until the pH of the solution is between 4.0 and 5.0.

c. Store primer in air-tight glass bottles or containers.

3.3 **Infrared identification**. The primer infrared identification shall conform qualitatively to that shown on figure 1.

3.4 **Color**. The primer shall be clear and colorless.

3.5 **Bond strength and durability**. The initial crack length (a_o) shall be less than 3.25 inches from the end of the plate and the crack extension (Δ_a) shall be less than 0.40 inches (average of 5 tests).

3.6 **Safety**.

3.6.1 **Toxic products and formulations**. The material shall have no adverse effect on the health of personnel when used for its intended purpose. Questions pertinent to this effect shall be referred by the contracting activity to the appropriate departmental medical service who will act as an advisor to the contracting agency (see 4.1.2).

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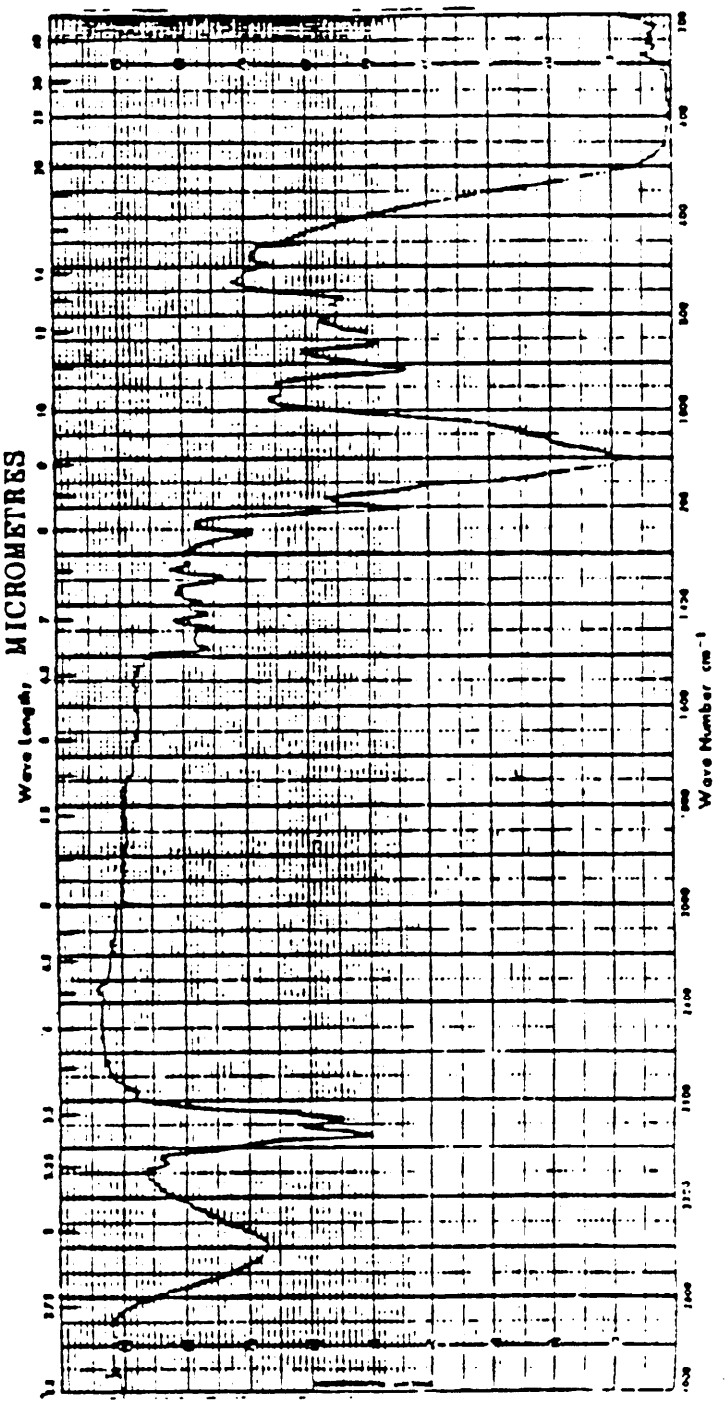


FIGURE 1. Primer infrared identification.

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3.6.2 **Material Safety Data Sheets (MSDS).** The contractor shall prepare and submit a MSDS in accordance with FED-STD-313 as specified in the contract (see 6.2).

3.7 **Shelf life.** The primer shall have a shelf life of 60 days from the date of manufacture when stored at 80 ± 20°F in the original closed containers.

3.8 **Workmanship.** The primer shall be free from agglomeration, foreign matter, separation, dilution, and contamination.

4. QUALITY ASSURANCE PROVISIONS

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 **Responsibility for compliance.** All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 **Toxicological product formulations.** The contractor shall have the toxicological product formulations and associated information available for review by the contracting activity to evaluate the safety of the material for the proposed use.

4.2 **Classification of inspections.** The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 **First article inspection.** First article inspection shall consist of the quality conformance inspections (see 4.4). Failure to meet all of the requirements specified herein shall be cause for rejection of the first article sample.

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4.4 **Quality conformance inspection.** Quality conformance inspection shall consist of the tests specified in table II.

TABLE II. Quality conformance inspection.

Tests and Examination	Requirement	Test Method
Visual examination	3.2, 3.4, 3.8 and section 5	4.5.1
IR identification	3.3	4.5.2
Bond strength and durability	3.5	4.5.3

4.4.1 **Lot.** A lot shall consist of all primer mixed in a single container at one time and subject to inspection at the same time.

4.4.2 **Sampling.** Sufficient samples shall be removed from each lot to conduct the tests specified herein.

4.4.3 **Classification of characteristics.** The characteristics verified by the inspections herein are classified as Major or Minor in accordance with WR-43. Inspections that verify Major characteristics are identified by the symbol (M). The number following the classification symbol indicates the serial number of the inspection. Inspections which are not annotated with a classification code and serial number are classified minor.

4.5 **Inspection methods.** The following inspection methods shall be used. Unless otherwise specified in the applicable test method, all weights, volumes, temperatures, and times shall be measured to the nearest specified unit or decimal.

NOTE: Reagent grade chemicals shall be used for chemical reactions in the conduct of all tests defined herein. Solvents and indicators may be commercial nonreagent grade materials.

4.5.1 **Visual examination.** Each container shall be visually examined to verify compliance with 3.4, 3.8 and section 5. Contractor certification verifying conformance to 3.2 shall be required (see 6.2).

4.5.2 **Infrared identification (M101).** The infrared identification of the primer shall be determined on a sample taken from each lot in accordance with the following:

a. Place a few drops of the primer on a sodium chloride plate. Spread to a thin coat.

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b. Allow to dry for 1 hour at room temperature or for 15 minutes at 120°F. Verify that the primer is dry.

c. Record infrared spectrum between 2.5 and 25 micrometers.

d. Compare the spectrum qualitatively with figure 1 and record as accept or reject.

4.5.3 Bond strength and durability. The procedure for wedge tests shall be in accordance with ASTM D 3762, with the following exceptions:

a. Prepare panels.

1. Cut 6 by 6 by 0.125 inch flat sheets of 4130 steel.

2. Vapor degrease sheets.

3. Grit blast faying surface with 100 to 200 zircon sand or equivalent at 100 ± 20 psi pressure.

4. Vapor degrease.

b. Assemble panels.

1. Cool panels to 75 ± 10°F after vapor degrease.

2. Apply primer as follows:

(a) Agitate primer bottle and pour the required amount of primer into a clean glass container.

(b) Paint primer onto faying surfaces using single brush strokes with minimal overlap.

(c) Do not return unused primer to the original container. Discard it.

(d) Dry primer for 1 hour minimum at 65 to 110°F or for 15 minutes at 120 ± 10°F.

3 Apply adhesive conforming to MIL-A-82845 to both faying surfaces.

4. Warm adhesive conforming to MIL-A-82845 (normally stored below room temperature) to within 10°F of ambient temperature before opening container. If necessary, place a small hole in the adhesive container for insertion of a thermometer. This will prevent moisture condensation on the adhesive and make the adhesive easier to spread.

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5. Apply FEP separation film in accordance with figure 1 of ASTM D 3762. Apply adhesive to both faying surfaces.

6. Mate faying surfaces and apply shrink tape to panel edges to prevent sliding.

7. Apply 1.0 ± 0.2 psi pressure uniformly on the panels during curing.

8. Cure at 250 ± 10°F for 2.00 ± 0.25 hours.

c. Cut panel into five 1-inch wide specimens.

1. Either a band saw or mill may be used. Apply liquid coolant during cutting operation.

2. Remove burrs, etc., from cut edges using a light grit blasting.

d. Insert wedge.

1. For steel adherends, the wedges shall be 0.059 ± 0.001 inch thick.

2. Wedges shall be inserted by initiating a debond (less than 1 inch) with a knife, then tapping the wedge into place with a hammer.

e. Mark initial crack length, a_o.

1. The initial crack length (a_o) shall be marked within 10 minutes after inserting the wedge.

f. Environment.

1. Subject specimens to 120 ± 5°F and 95 to 100 percent relative humidity for 3.0 ± 0.5 hours.

g. Crack extension, Δ_a, will be recorded only at 3 ± 0.5 hours.

h. Record average initial crack length (a_o) and crack extension (Δ_a).

4.6 **Inspection of packaging.** The primer containers and container markings shall be visually examined to verify compliance with section 5.

5. PACKAGING

5.1 **Packaging.** Unless otherwise specified in the contract or order (see 6.2), packaging shall be level C as specified herein.

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5.1.1 **Level C.** The primer shall be packaged to afford adequate protection against loss, contamination, deterioration, and damage during shipment from the supply source to the first receiving activity and during storage under the shelf life period and conditions specified in 3.7. Containers in the same shipment shall be of the same size. The packaging shall conform to UFC 6000, National Motor Freight Classification, 49 CFR 171-178, or to other carrier rules and regulations as applicable to the mode of transportation.

5.2 **Marking.**

5.2.1 **Standard marking.** In addition to any special marking required by the contract or order (see 6.2), interior and exterior containers shall be marked in accordance with MIL-STD-129.

5.2.2 **Special marking.** In addition to the marking requirements of 5.2.1, each container marking shall include the following:

a. Title, number, and date of this specification.

b. Manufacturer's name and address.

c. Date of manufacture.

d. Shelf life expiration date.

e. Storage conditions.

f. Toxicity warning (if applicable).

6. **NOTES**

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 **Intended use.** The primer is intended for use in the manufacture of nozzle components for the Standard Missile Mk 104 Dual Thrust Rocket Motor.

6.2 **Acquisition requirements.** Acquisition documents must specify the following:

a. Title, number and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).

c. Whether first article inspection is required (see 3.1 and 6.3).

d. Material Safety Data Sheets Required (see 3.6.2 and 6.6).

e. DOD FAR 52.246-15 (see 4.5.1).

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f. Special marking, if other than as specified in 5.2.2.

g. Packaging required, if other than as specified in 5.1.1.

6.3 **First article.** When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first production items, or a standard production item from the contractor's current inventory (see 3.1), and the number of items to be tested as specified in 4.3. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 **Possible material.** The following material has been found suitable for use as an ingredient in the primer. This is only given for information and is not restrictive. Union Carbide A-187 has been found satisfactory as an ingredient.

6.5 **Material Safety Data Sheets.** MSDS requirements are applicable to this specification and should be specified in the contract as required by the Federal Acquisition Regulation (FAR) Part 23, subpart 23.3. Contracting officers will identify those activities requiring copies of completed Material Safety Data Sheets prepared in accordance with FED-STD-313. The pertinent Government mailing addresses for submission of data are listed in FED-STD-313.

6.6 **Subject term (key word) listing.**

Nozzle, rocket
Rocket Motor, Dual Thrust, Mk 104
Standard Missile

6.7 **Supersedure information.** MIL-P-82816 incorporates the following engineering change proposal (ECP) and specification change notice (SCN):

ECP		SCN	
MTA058	(12/14/89)	SCN 1	(12/21/87)

Preparing Activity:
NAVY-OS
(Project 8040-N149)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-P-82816	2. DOCUMENT DATE (YYMMDD) 5 February 1990

3. DOCUMENT TITLE
PRIMER, ADHESIVE

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY NAVAL ORDNANCE STATION (CODE 3730) INDIAN HEAD, MD 20640-5000

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON
c. ADDRESS (Include Zip Code)	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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